SIMATS ENGINEERING

ASSESSMENT TOOL 1 | ANNEXURE B

ANALYTICAL PROBLEM SOLVING

Data Preprocessing: Covariance, Binning-Based Smoothing, Discretisation and Standardisation

Course Code	CSA16
Course Name	Data Warehousing and Data Mining
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Register Number	192421295
Course Outcome	CO2 – Apply suitable data preprocessing techniques (data cleaning, integration, transformation and reduction) for effective data preparation (BL3, BL4)
Assessment Weightage	50%
Number of Questions / Marks	5 questions | Total 50 marks
Date of Submission	23 October 2026

Code of Conduct

I, Manoj A (Reg. No. 192421295), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Manoj A

How this report is organised

Every question is answered in the five stages used by the marking rubric, so each criterion can be located quickly:

- **Problem Understanding** – what is given, what is asked, and the parameters involved.
- **Method Selection** – the formula or technique chosen and why it is the most appropriate.
- **Solution Process** – complete, step-by-step working with intermediate tables.
- **Final Result** – the answer, cross-checked by a second route wherever possible.
- **Interpretation** – what the numbers mean in the healthcare / insurance context, with units and limitations.

Question 1 – Covariance Analysis in Health Insurance (10 Marks)

1.1 Problem Understanding

A health insurer has recorded the **annual premium increase rate** (x_i, in %) and the **number of claims per 100 policyholders** (y_i) for four observations. The task is to decide, using the covariance formula, whether the two variables move together (positive relationship) or in opposite directions (inverse relationship). The question also requires that the means of x and y be calculated first.

Observation (i)	Premium increase, x_i (%)	Claims per 100 policyholders, y_i
1	3.2	15
2	4.0	18
3	5.5	25
4	4.8	20

- Number of paired observations: **n = 4**.
- Both attributes are numeric (ratio-scaled); x is treated as the explanatory variable and y as the response.
- Required output: sign of Cov(x, y) and a statement on direct / inverse association.

1.2 Method Selection

Covariance measures the joint variability of two attributes. A positive value means that when one attribute is above its mean the other tends to be above its mean as well; a negative value means they move in opposite directions; a value near zero indicates no linear association. The defining formula (as used in data-integration redundancy analysis) is:

$$Cov(x, y) = (1/n)\,\Sigma\,(x_i - \bar{x})(y_i - \bar{y}) \;=\; E(x{\cdot}y) - \bar{x}{\cdot}\bar{y}$$

The deviation form is chosen for the main calculation because it shows each pair's contribution to the sign of the result, and the computational form $E(x{\cdot}y) - \bar{x}\bar{y}$ is used as an independent check. The sample version (divide by $n - 1$) is also reported, since both conventions appear in practice. Because covariance depends on the units of x and y, the Pearson correlation coefficient $r = Cov / (\sigma_x \sigma_y)$ is added as a unit-free confirmation.

1.3 Solution Process

Step 1 – Compute the means

$$\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 17.5 / 4 = 4.375\ \%$$

$$\bar{y} = (15 + 18 + 25 + 20) / 4 = 78 / 4 = 19.5\ claims\ per\ 100\ policyholders$$

Step 2 – Deviations from the means and their products

i	x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
1	3.2	15	−1.175	−4.5	5.2875

i	x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
2	4.0	18	−0.375	−1.5	0.5625
3	5.5	25	+1.125	+5.5	6.1875
4	4.8	20	+0.425	+0.5	0.2125
Σ	**17.5**	**78**	**0.000**	**0.0**	**12.2500**

The deviation columns sum to zero, which confirms that the means were computed correctly. Pairs 1 and 2 lie below both means (negative × negative = positive product) and pairs 3 and 4 lie above both means (positive × positive = positive product), so **every product is positive**.

Step 3 – Covariance

$$Cov(x, y) = 12.25 / 4 = 3.0625 \quad \textit{(population form)}$$

$$Cov\text{⍰}(x, y) = 12.25 / (4 - 1) = 4.0833 \quad \textit{(sample form)}$$

Step 4 – Verification using $E(x \cdot y) - \bar{x}\bar{y}$

i	x_i	y_i	$x_i \times y_i$	Working
1	3.2	15	48.0	3.2 × 15
2	4.0	18	72.0	4.0 × 18
3	5.5	25	137.5	5.5 × 25
4	4.8	20	96.0	4.8 × 20
Σ			**353.5**	

$$E(x \cdot y) = 353.5 / 4 = 88.375; \quad \bar{x}\bar{y} = 4.375 \times 19.5 = 85.3125$$

$$Cov(x, y) = 88.375 - 85.3125 = 3.0625 \quad \checkmark \textit{ matches Step 3}$$

Step 5 – Unit-free confirmation (correlation) and trend line

Quantity	Working	Value
$\Sigma(x_i - \bar{x})^2$	1.380625 + 0.140625 + 1.265625 + 0.180625	2.9675
σ_x^2 (population)	2.9675 / 4	0.741875 (σ_x = 0.8613)
$\Sigma(y_i - \bar{y})^2$	20.25 + 2.25 + 30.25 + 0.25	53.00
σ_Y^2 (population)	53 / 4	13.25 (σ_Y = 3.6401)
Correlation r	3.0625 / (0.8613 × 3.6401)	0.9768
Slope b = Cov / σ_x^2	3.0625 / 0.741875	4.128 claims per 1 pp
Intercept a = $\bar{y} - b \cdot \bar{x}$	19.5 − 4.128 × 4.375	1.440

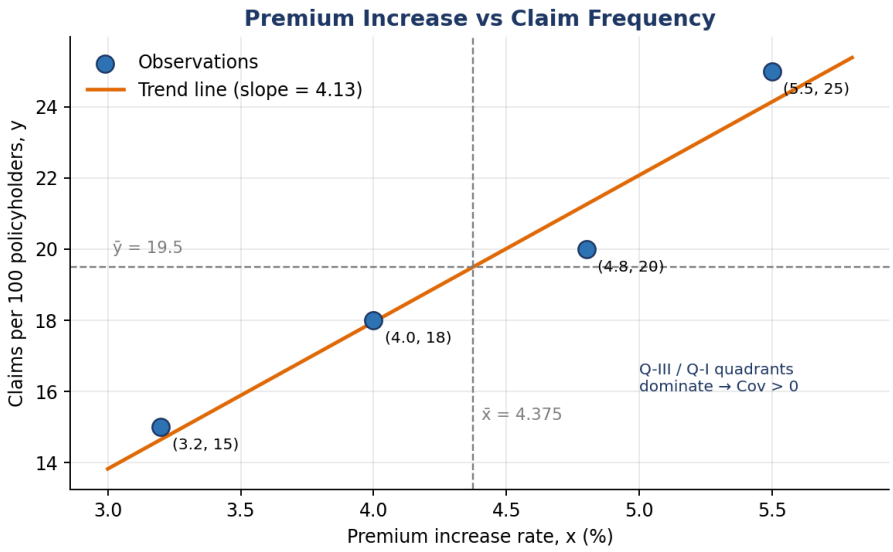


Figure 1: Scatter plot of premium increase versus claim frequency with fitted trend line

1.4 Final Result

Measure	Result
Mean of x (premium increase)	$\bar{x}$ = 4.375 %
Mean of y (claims per 100 policyholders)	$\bar{y}$ = 19.5
Covariance (population)	Cov(x, y) = +3.0625
Covariance (sample)	Cov□(x, y) = +4.0833
Correlation coefficient	r ≈ +0.977
Conclusion	**POSITIVE (direct) relationship**

1.5 Interpretation of Results

- **Direction.** Cov(x, y) > 0, so premium increase rate and claim frequency have a **positive (direct) relationship, not an inverse one**: years in which the premium rose more were also years in which more claims were filed per 100 policyholders.
- **Strength.** The covariance itself has no natural scale (its unit is % × claims), but the standardised value r ≈ 0.977 shows the association is close to perfectly linear. Each additional percentage point of premium increase corresponds to roughly 4.1 extra claims per 100 policyholders.
- **Business meaning.** The pattern is consistent with insurers raising premiums in response to rising claims experience (the usual actuarial feedback loop), and may also reflect adverse selection, where higher premiums push healthier members out and leave a riskier pool.
- **Data-mining relevance.** In data integration, two attributes with such a high positive correlation are candidates for redundancy removal; keeping both adds little new information to a mining model.
- **Limitations.** Only four observations are available, so the estimate is unstable, and association does not prove causation. A larger, multi-year dataset should be used before any pricing decision.

Question 2 – Smoothing by Equal-Frequency (Equi-depth) Binning (5 Marks)

2.1 Problem Understanding

A hospital wants to remove minor fluctuations from 27 systolic blood-pressure readings so that underlying trends are easier to see. The technique requested is **local smoothing using equal-frequency binning**: the sorted data are divided into bins that each contain the same number of values, and the values inside each bin are then replaced by a representative value.

Parameter	Value
Attribute	Systolic blood pressure (mmHg), already sorted ascending
Number of values, n	27
Number of bins, k	3 (assumed – not specified in the question; 27 divides evenly)
Bin depth (values per bin)	27 / 3 = 9

Data: 110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

2.2 Method Selection

Equal-frequency partitioning is the right choice here because blood-pressure readings are unevenly spread (many values near 120–140 and a few high readings). Equal-**width** bins would put almost all readings in one or two bins, whereas equal-**depth** bins guarantee that each bin carries the same amount of data and therefore gives good handling of skewed values. Smoothing is then performed **by bin means**, the standard local-smoothing choice, with bin medians and bin boundaries shown as alternatives for comparison.

$$Depth = n / k = 27 / 3 = 9 \qquad Bin\ mean = (\Sigma\ values\ in\ bin) / 9$$

2.3 Solution Process

Step 1 – Partition the sorted data into equal-depth bins

Bin	Values (9 each)	Σ	Mean	Median
Bin 1	110, 112, 115, 115, 118, 120, 120, 122, 124	1056	117.33	118
Bin 2	124, 126, 126, 126, 126, 130, 135, 135, 138	1166	129.56	126
Bin 3	138, 138, 138, 140, 145, 150, 152, 160, 180	1341	149.00	145

The values 124 and 138 occur in two neighbouring bins because ties are split by position: an equal-frequency scheme counts values, not distinct numbers, so exactly nine readings go into every bin.

Step 2 – Bin means

$$Bin\ 1: 1056 / 9 = 117.33 \qquad Bin\ 2: 1166 / 9 = 129.56 \qquad Bin\ 3: 1341 / 9 = 149.00$$

Step 3 – Smoothing by bin means (main answer)

Bin	Smoothed values
Bin 1	117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33
Bin 2	129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56
Bin 3	149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00

Step 4 – Alternative smoothing (for comparison)

By bin boundaries – each value is replaced by the nearer of the bin's minimum and maximum (e.g. in Bin 1, 118 is 8 from 110 but only 6 from 124, so it becomes 124):

Bin	Smoothed values (nearest boundary)
Bin 1	110, 110, 110, 110, 124, 124, 124, 124, 124 (boundaries 110 & 124)
Bin 2	124, 124, 124, 124, 124, 124, 138, 138, 138 (boundaries 124 & 138)
Bin 3	138, 138, 138, 138, 138, 138, 138, 180, 180 (boundaries 138 & 180)

By bin medians – Bin 1 → 118 (nine times), Bin 2 → 126 (nine times), Bin 3 → 145 (nine times).

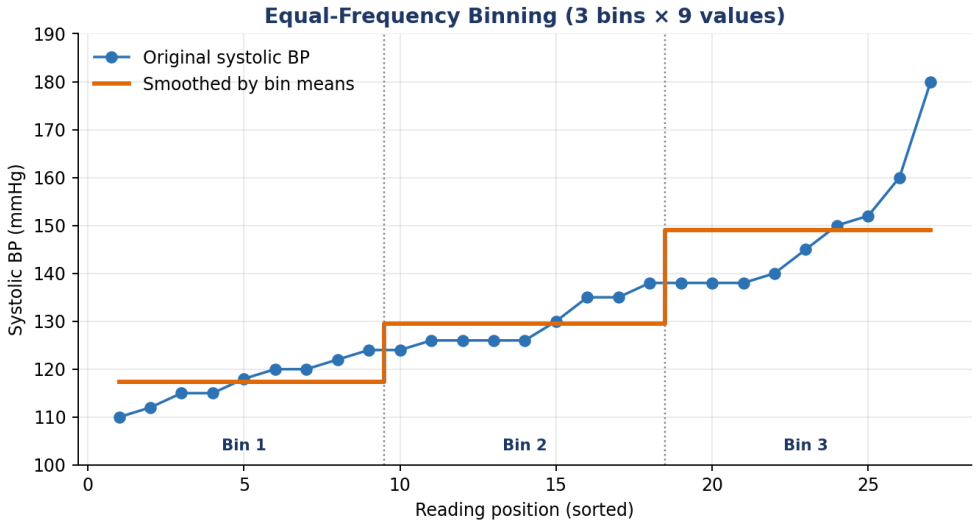


Figure 2: Original systolic readings and the bin-mean smoothed series

2.4 Final Result

Bin	Range of original values	Smoothed by mean	By median	By boundaries
Bin 1	110 – 124	117.33	118	110 / 124
Bin 2	124 – 138	129.56	126	124 / 138
Bin 3	138 – 180	149.00	145	138 / 180

2.5 Interpretation of Results

- **Trend revealed.** The 27 fluctuating readings collapse into three clear levels – about **117, 130 and 149 mmHg** – showing a steady upward gradient across the patient set that was hard to see in the raw sequence.
- **Clinical reading.** Using common hypertension categories, the Bin 1 level is in the normal range, Bin 2 in the elevated / stage-1 border zone, and Bin 3 in the stage-2 hypertension range.
- **Outlier effect.** Bin 3's mean (149) is noticeably above its median (145) because of the single 180 mmHg reading. Mean smoothing therefore lets one extreme value pull the whole bin upward; median or boundary smoothing is more robust when outliers are present.
- **Caution.** A reading of 180 mmHg sits at the hypertensive-crisis threshold. Smoothing is appropriate for trend analysis and mining, but the original value must be retained for patient care so that this clinically important reading is not hidden.

Question 3 – Equal-Width Binning for BMI Risk Categorisation (5 Marks)

3.1 Problem Understanding

An insurer has the sorted Body Mass Index (BMI) values of 18 clients. To simplify risk analysis, the values must be **smoothed by replacing each one with its closest bin boundary**, where the bins are produced by **equal-width partitioning**.

Data: 18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Parameter	Value
Number of values, n	18
Minimum (A) and maximum (B)	A = 18.5, B = 39.5
Number of bins, k	3 (assumed – the question does not fix k; three bands map naturally onto low / medium / high risk)

3.2 Method Selection

In equal-width partitioning the value range is divided into k intervals of identical size, regardless of how many values fall in each. The width is:

$$W = (B - A) / k = (39.5 - 18.5) / 3 = 7.0$$

Interval limits therefore fall at 18.5, 25.5, 32.5 and 39.5. Each interval is taken as [lower, upper), except that the last interval also includes its upper limit, so 25.5 belongs to Bin 2. **Smoothing by bin boundaries** then replaces every value by whichever of the bin's two boundary values (its minimum or maximum) is nearer; in the event of a tie the lower boundary would be used.

3.3 Solution Process

Step 1 – Assign values to bins

Bin	Interval	Count	Values	Min / Max
Bin 1	[18.5, 25.5)	4	18.5, 19.2, 21.8, 24.9	18.5 / 24.9
Bin 2	[25.5, 32.5)	9	25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4	25.5 / 32.4
Bin 3	[32.5, 39.5]	5	33.1, 33.6, 34.7, 38.2, 39.5	33.1 / 39.5

Check: 4 + 9 + 5 = 18 values. The counts are unequal – this is the defining difference between equal-width and equal-frequency binning.

Step 2 – Distance of every value to its bin boundaries

Bin	Value	Distance to min	Distance to max	Closer boundary	Smoothed value
1	18.5	0.0	6.4	min	18.5
1	19.2	0.7	5.7	min	18.5

Bin	Value	Distance to min	Distance to max	Closer boundary	Smoothed value
1	21.8	3.3	3.1	max	24.9
1	24.9	6.4	0.0	max	24.9
2	25.5	0.0	6.9	min	25.5
2	26.2	0.7	6.2	min	25.5
2	26.4	0.9	6.0	min	25.5
2	27.8	2.3	4.6	min	25.5
2	29.2	3.7	3.2	max	32.4
2	30.2	4.7	2.2	max	32.4
2	30.4	4.9	2.0	max	32.4
2	31.9	6.4	0.5	max	32.4
2	32.4	6.9	0.0	max	32.4
3	33.1	0.0	6.4	min	33.1
3	33.6	0.5	5.9	min	33.1
3	34.7	1.6	4.8	min	33.1
3	38.2	5.1	1.3	max	39.5
3	39.5	6.4	0.0	max	39.5

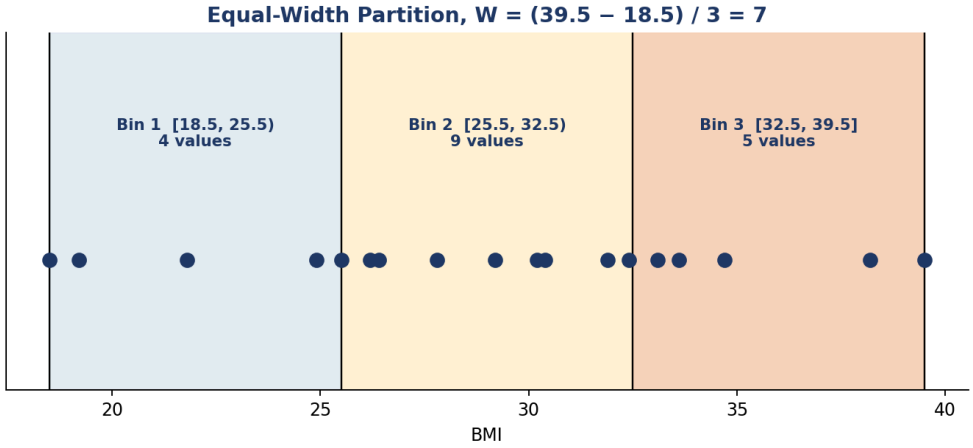


Figure 3: Equal-width partition of the 18 BMI values into three bins of width 7

Step 3 – Smoothed data set

Bin	Smoothed values (by bin boundaries)
Bin 1	18.5, 18.5, 24.9, 24.9
Bin 2	25.5, 25.5, 25.5, 25.5, 32.4, 32.4, 32.4, 32.4, 32.4
Bin 3	33.1, 33.1, 33.1, 39.5, 39.5

Cross-check – if "bin boundary" is read as the partition limit

Some textbooks use the interval limits (18.5, 25.5, 32.5, 39.5) rather than the smallest and largest values actually present. Applying that reading gives the following result, so either interpretation is covered:

Bin	Smoothed values (nearest interval limit)
Bin 1	18.5, 18.5, 18.5, 25.5 (limits 18.5 & 25.5; 21.8 is 3.3 from 18.5 but 3.7 from 25.5)
Bin 2	25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5 (limits 25.5 & 32.5)
Bin 3	32.5, 32.5, 32.5, 39.5, 39.5 (limits 32.5 & 39.5)

3.4 Final Result

Bin	Interval (W = 7)	Smoothed values (closest boundary)
Bin 1	18.5 – 25.5	18.5, 18.5, 24.9, 24.9
Bin 2	25.5 – 32.5	25.5, 25.5, 25.5, 25.5, 32.4, 32.4, 32.4, 32.4, 32.4
Bin 3	32.5 – 39.5	33.1, 33.1, 33.1, 39.5, 39.5

3.5 Interpretation of Results

- **Risk bands.** The three bins correspond approximately to a **lower-risk band** (BMI about 18.5–25, the WHO normal range), a **moderate-risk band** (about 25.5–32.4, covering overweight and the start of obesity) and a **higher-risk band** (33 and above, obesity class I–II). The smoothed values let an underwriter apply one loading per band instead of 18 individual BMI figures.
- **Uneven population.** Nine of the 18 clients (50 %) fall in the middle band, only four in the lowest. Equal-width bins do not balance counts, but they do keep the BMI scale uniform, which is what makes the bands easy to explain to policyholders.
- **Information loss.** Boundary smoothing pushes values to the extremes of their bin (21.8 moves up to 24.9, 29.2 moves up to 32.4), so it slightly overstates risk for mid-bin clients. If finer pricing is needed, more bins (e.g. k = 4 or 5) should be used.

Question 4 – Data Reduction Using Bin Medians (5 Marks)

4.1 Problem Understanding

A hospital administration holds 27 cost records for medical consumables (in ₹). To reduce the volume of distinct values before mining, the sorted records are to be **discretised by replacing each value with the median of its bin**.

Data as given (not sorted): 120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

- The question speaks of "sorted price records" but the list is **not in order** (for example, 140 appears after 160), so sorting is the compulsory first step.
- n = 27 records, 16 distinct values. Number of bins k = 3 is assumed (equal depth of 9), the same convention used in Question 2.

4.2 Method Selection

Binning with medians is chosen because hospital costs are right-skewed: most items cost ₹140–₹180 but a few (₹230, ₹300) are far higher. The **median** of a bin is not affected by such extremes, so it represents the typical item better than the mean. Equal-frequency bins are used so that each cost tier covers the same number of records.

$$\textit{Bin depth} = 27 / 3 = 9; \quad \textit{median} = 5\textit{th value of each sorted bin of } 9$$

4.3 Solution Process

Step 1 – Sort the data

120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Step 2 – Form three bins of nine values and locate each median

Bin	Values (sorted)	5th value (median)	Mean	Range
Bin 1	120, 140, 145, 145, 148, 150, 150, 150, 155	₹148	₹144.78	120–155
Bin 2	155, 155, 155, 160, 160, 165, 170, 170, 175	₹160	₹162.78	155–175
Bin 3	175, 175, 175, 180, 190, 200, 210, 230, 300	₹190	₹203.89	175–300

With nine values per bin the median is the middle (5th) value: Bin 1 → 148, Bin 2 → 160, Bin 3 → 190.

Step 3 – Replace every value with its bin median

Bin	Discretised (reduced) values
Bin 1	148, 148, 148, 148, 148, 148, 148, 148, 148
Bin 2	160, 160, 160, 160, 160, 160, 160, 160, 160

Bin	Discretised (reduced) values
Bin 3	190, 190, 190, 190, 190, 190, 190, 190, 190

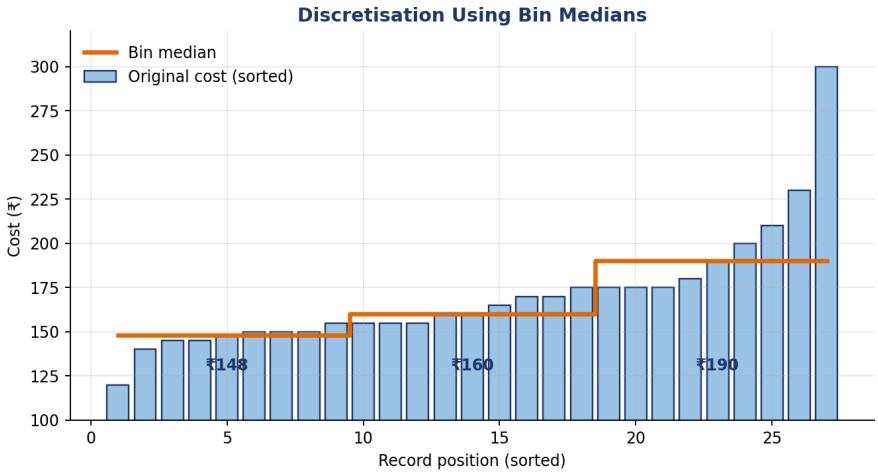


Figure 4: Sorted consumable costs and the three bin-median levels

Step 4 – Measure the reduction and the information loss

Bin	Σ |v − median|	Σ |v − mean|	Median	Mean
Bin 1	55	59.11	148	144.78
Bin 2	55	57.78	160	162.78
Bin 3	235	256.67	190	203.89
Total	**345**	**373.56**		

The median gives a smaller total absolute error (345 against 373.56), which confirms it is the better single representative for this skewed data – a median minimises the sum of absolute deviations.

Measure	Before	After
Records	27	27
Distinct cost values	16	3
Reduction in distinct values	–	(16 − 3) / 16 = 81.25 %

4.4 Final Result

Bin	Original values	Replaced by
Bin 1 (low cost)	₹120 – ₹155	₹148
Bin 2 (medium cost)	₹155 – ₹175	₹160
Bin 3 (high cost)	₹175 – ₹300	₹190

4.5 Interpretation of Results

- **Reduction achieved.** 16 distinct prices are compressed into 3 representative tiers (₹148, ₹160, ₹190), an 81.25 % reduction in distinct values, which makes frequency counting, rule mining and cost-category reports much lighter.
- **Why median, not mean.** Bin 3's mean would be ₹203.89 because of the ₹300 item, overstating the typical high-tier cost by about ₹13.89 compared with the median of ₹190. The median keeps the tier representative of ordinary purchases.
- **Budgeting use.** The hospital can plan consumable budgets on three standard price points instead of individual prices.
- **Limitation.** The ₹300 purchase becomes ₹190 after discretisation, so unusually expensive items are masked. For procurement audits the original records should be kept alongside the reduced attribute.

Question 5 – Standardisation of Patient Age Data (5 Marks)

5.1 Problem Understanding

The ages (in years) of ten insured individuals are given. The variable must be standardised so that it is on a unit-free scale suitable for mining algorithms. Two items are required: (a) the **mean absolute deviation (MAD)** of age, and (b) the **z-score of the first four observations**.

Data: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28 (n = 10)

5.2 Method Selection

Because part (a) asks for the MAD before part (b) asks for the z-scores, the standardisation is the **robust form of the z-score**, in which the mean absolute deviation replaces the standard deviation as the measure of spread. This is preferred in data preprocessing because absolute deviations are not squared, so an unusually old or young customer has less influence on the scale than it would on the standard deviation.

$$MAD\ (s_a) = (1/n)\ \Sigma\ |\ x_i - \bar{x}\ |\qquad z_i = (x_i - \bar{x})\ /\ s_a$$

For comparison, the conventional z-score using the sample standard deviation is also shown.

5.3 Solution Process

Step 1 – Mean age

$$\bar{x} = (18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28)\ /\ 10 = 330\ /\ 10 = 33\ years$$

Step 2 – Absolute deviations and MAD (part a)

i	Age x_i	$x_i - \bar{x}$	$\lvert x_i - \bar{x} \rvert$	$(x_i - \bar{x})^2$	Note
1	18	−15	15	225	
2	22	−11	11	121	
3	25	−8	8	64	
4	42	+9	9	81	
5	28	−5	5	25	
6	43	+10	10	100	
7	33	0	0	0	equals mean
8	35	+2	2	4	
9	56	+23	23	529	largest
10	28	−5	5	25	
Σ	**330**	**0**	**88**	**1174**	

The signed deviations add to zero, confirming the mean. Therefore:

$$MAD = 88\ /\ 10 = 8.8\ years$$

For the comparison scale: sample variance = 1174 / 9 = 130.44, so the sample standard deviation s = √130.44 = 11.42 years.

Step 3 – z-scores of the first four observations (part b)

Obs.	Age	z = (x − 33) / 8.8	z (MAD)	z (using s = 11.42)
1	18	−15 / 8.8	−1.70	−1.31
2	22	−11 / 8.8	−1.25	−0.96
3	25	−8 / 8.8	−0.91	−0.70
4	42	+9 / 8.8	+1.02	+0.79

Exact values: −15/8.8 = −1.7045; −11/8.8 = −1.2500; −8/8.8 = −0.9091; +9/8.8 = +1.0227.

Extension – z-scores of the remaining observations

Obs.	Age	z (MAD)	z (s = 11.42)	Comment
5	28	−0.57	−0.44	
6	43	+1.14	+0.88	
7	33	0.00	0.00	at the mean
8	35	+0.23	+0.18	
9	56	+2.61	+2.01	most extreme
10	28	−0.57	−0.44	

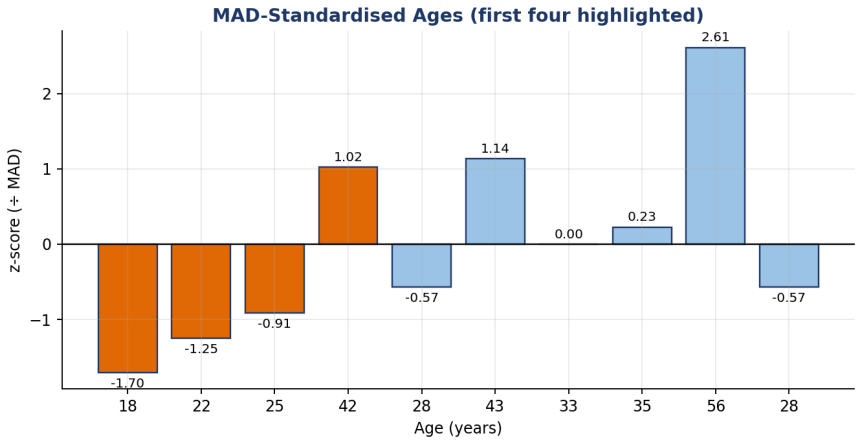


Figure 5: MAD-based z-scores for all ten ages (first four observations highlighted)

5.4 Final Result

Item	Answer
(a) Mean age	33 years
(a) Mean absolute deviation	**MAD = 8.8 years**
(b) z-score, age 18	−1.70 (−1.7045)

Item	Answer
(b) z-score, age 22	−1.25
(b) z-score, age 25	−0.91 (−0.9091)
(b) z-score, age 42	+1.02 (+1.0227)

5.5 Interpretation of Results

- **Meaning of the scores.** The 18-year-old is 1.70 MAD-units **below** the average age of 33, the 22-year-old is 1.25 units below, the 25-year-old 0.91 units below, and the 42-year-old 1.02 units **above**. A z-score has no units, so ages can now be combined with attributes such as premium amount or claim count in distance-based mining (k-means, k-nearest-neighbour) without age dominating because of its scale.
- **Outliers.** None of the first four observations exceeds $|z| = 2$, so they are ordinary members of the group. The 56-year-old ($z = +2.61$) is the only value beyond 2 MAD-units and is the one candidate for outlier review.
- **MAD versus standard deviation.** MAD (8.8) is smaller than the standard deviation (11.42) because squaring inflates the effect of the 56-year-old. MAD-based scores are therefore larger in magnitude but less distorted by that extreme value.
- **Property check.** The standardised variable has mean 0 by construction, since the signed deviations sum to zero.

Summary of Final Answers

Q.	Technique	Final answer
1	Covariance	$\bar{x}$ = 4.375, $\bar{y}$ = 19.5; Cov(x, y) = +3.0625 (sample +4.0833); r ≈ 0.977 → positive (direct) relationship
2	Equal-frequency binning (smoothing by means)	Bin means 117.33, 129.56, 149.00 (9 values each)
3	Equal-width binning (smoothing by boundaries)	W = 7; bins of 4, 9, 5 values; smoothed to 18.5/24.9, 25.5/32.4, 33.1/39.5
4	Discretisation by bin medians	Medians ₹148, ₹160, ₹190; distinct values reduced from 16 to 3
5	Standardisation	Mean = 33; MAD = 8.8; z = −1.70, −1.25, −0.91, +1.02

Assumptions Used

- Covariance is reported in both population (÷ n) and sample (÷ n − 1) forms; the population form is the primary answer.
- Where the number of bins was not stated (Questions 2, 3 and 4), k = 3 was used; for Questions 2 and 4 this gives an exact depth of 9 values per bin.
- In Question 3 the interval convention is [lower, upper) with the last interval closed.
- Standardisation in Question 5 divides by the mean absolute deviation, as required by parts (a) and (b); the standard-deviation version is shown for comparison.
- Intermediate values are held at full precision and rounded to two decimals only for presentation.

References

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